

Problem

Sketch the waveform represented by

$$i(t) = [r(t) - r(t-1) - u(t-2) - r(t-2) + r(t-3) + u(t-4)] \text{ A}$$

Step-by-step solution

Step 1 of 9

Consider the following current signal $i(t)$.

$$i(t) = r(t) - r(t-1) - u(t-2) - r(t-2) + r(t-3) + u(t-4)$$

Split the signal $i(t)$ as,

$$i(t) = i_1(t) + i_2(t)$$

Here,

$$i_1(t) = r(t) - r(t-1) - r(t-2) + r(t-3)$$

$$i_2(t) = -u(t-2) + u(t-4)$$

Step 2 of 9

Consider the current $i_1(t)$.

Define the following signals:

$$r(t) = \begin{cases} t, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

$$-r(t-1) = \begin{cases} -t, & t > 1 \\ 0, & t \leq 1 \end{cases}$$

$$-r(t-2) = \begin{cases} -t, & t > 2 \\ 0, & t \leq 2 \end{cases}$$

$$r(t-3) = \begin{cases} t, & t > 3 \\ 0, & t \leq 3 \end{cases}$$

Step 3 of 9

Sketch the waveform of $r(t)$, $-r(t-1)$, $-r(t-2)$, and $r(t-3)$ in a single plot as shown in Figure 1.

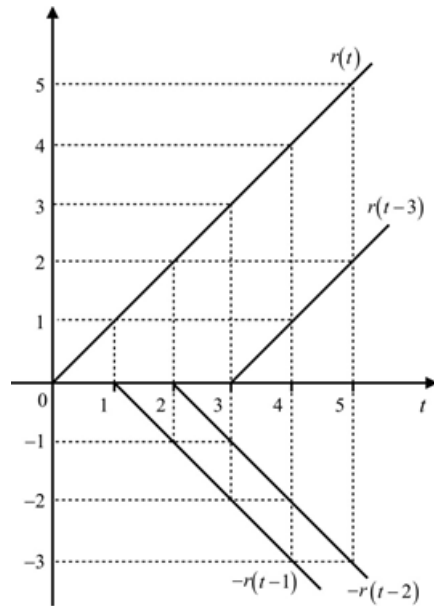


Figure 1

Step 4 of 9

Define the current $i_1(t)$.

Add the signals that were plotted in Figure 1.

$$i_1(t) = \begin{cases} t, & 0 < t < 1 \\ 1, & 1 < t < 2 \\ -t+3, & 2 < t < 3 \\ 0, & t > 3 \end{cases}$$

Step 5 of 9

Sketch the waveform of current $i_1(t)$ as shown in Figure 2.

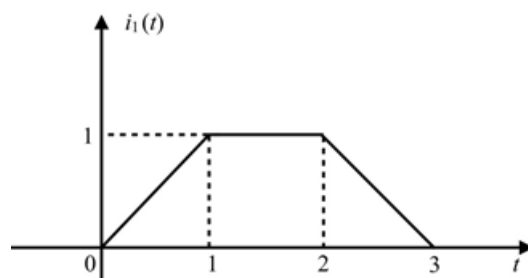


Figure 2

Step 6 of 9

Consider the current $i_2(t)$.

Define the following signals:

$$-u(t-2) = \begin{cases} -1, & t > 2 \\ 0, & t \leq 2 \end{cases}$$

$$u(t-4) = \begin{cases} 1, & t > 4 \\ 0, & t \leq 4 \end{cases}$$

Sketch the waveform of $-u(t-2)$ and $u(t-4)$ in a single plot as shown in Figure 3.

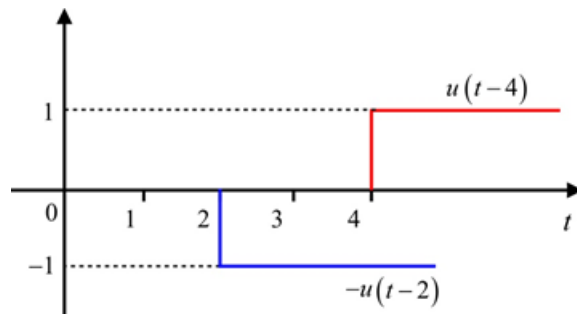


Figure 3

Step 7 of 9

Define the current $i_2(t)$.

Add the signals that were plotted in figure 3.

$$i_2(t) = \begin{cases} -1, & 2 < t < 4 \\ 0, & \text{otherwise} \end{cases}$$

Sketch the waveform of current $i_2(t)$ as shown in Figure 4.

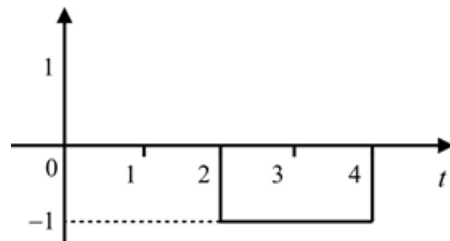


Figure 4

Step 8 of 9

Define current $i(t)$.

Consider the waveforms shown in Figures 2 and 4.

Add the currents $i_1(t)$ and $i_2(t)$.

$$\begin{aligned} i(t) &= i_1(t) + i_2(t) \\ &= \begin{cases} r(t) & 0 \leq t \leq 1 \\ 1 & 1 \leq t \leq 2 \\ -r(t) & 2 \leq t \leq 3 \\ -1 & 3 \leq t \leq 4 \end{cases} \end{aligned}$$

Step 9 of 9

Plot the waveform of current $i(t)$ as shown in Figure 5.

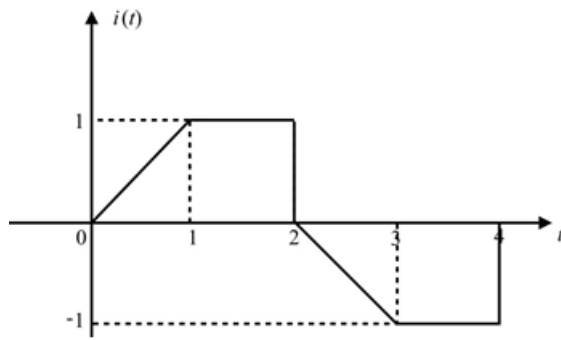


Figure 5

Therefore, the waveform of current $i(t)$ is shown in Figure 5.